

4. ROS2 Workspace

Note: The accompanying reference video uses the ROS Humble version, for reference only

1. Workspace Overview

In ROS robot development, when developing specific functions for a robot, all your code, parameters, scripts, and other files need to be placed in a folder for management. This folder is called the workspace in the ROS system. A workspace is a folder that stores project development-related files, serving as the base camp for storing all materials during the development process.

2. Creating a Workspace

- Create a folder as the workspace for storing project files. yahboomcar_ws is the folder name, you can name it arbitrarily.

```
mkdir -p yahboomcar_ws/src
```

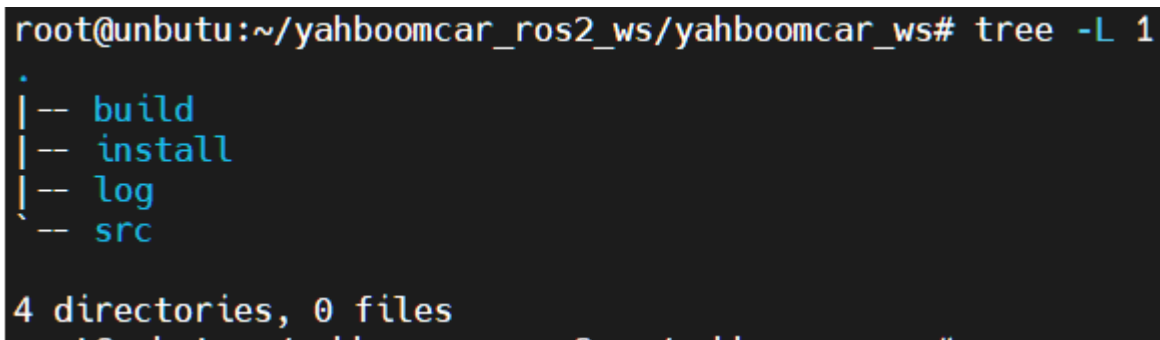
- Navigate to the workspace folder:

```
cd ~/yahboomcar_ws
```

3. Building the Workspace

```
colcon build
```

After the build completes, you should see the `build`, `install`, and `log` directories:



```
root@ubuntu:~/yahboomcar_ros2_ws/yahboomcar_ws# tree -L 1
.
|-- build
|-- install
|-- log
`-- src

4 directories, 0 files
```

A typical workspace structure in ROS is shown above. yahboomcar_ws is the root directory of the workspace, containing four subdirectories, also called four subspaces.

- **src, Source Space:** Where future code and scripts you write are manually placed;
- **build, Build Space:** Stores intermediate files generated during the build process;
- **install, Install Space:** Contains compiled executables and scripts;

- **log, Log Space:** Stores various warnings, errors, and information logs during build and runtime.

Generally speaking, most of our operations are performed in the src directory. After successful build, the results in the install directory are executed. The build and log directories are rarely used.

It should also be emphasized that **the workspace name can be defined by ourselves**, and the number of workspaces is not limited. For example:

```
workspace 1: ros2_ws_a, for Robot A development
workspace 2: ros2_ws_b, for Robot B development
workspace 3: ros2_ws_c, for Robot C development
```

The above is completely acceptable, just like creating multiple new projects in an integrated development environment. They coexist as 并列关系.

4. Setting Up Environment Variables

After successful build, to enable the system to find your packages and executables, you need to set environment variables:

```
# Only takes effect in current terminal
source install/setup.bash
# Takes effect in all terminals
echo "source ~/yahboomcar_ws/install/setup.bash" >> ~/.bashrc
```

At this point, the creation, building, and configuration of the workspace is complete.